Task 1 - QA Report: RealWorld (Conduit) app

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Scope & setup. demo.realworld.io was offline (404 NoSuchBucket), so I ran the app locally: FastAPI + PostgreSQL backend (realworld-fastapi) on :8000 and the React-Redux frontend on :4100. Testing was manual exploratory testing plus scripted Playwright flows (sign-up, login, editor, delete, logout) and direct API calls. Areas covered: auth, article CRUD, validation, session handling, error display, basic accessibility and security. Evidence screenshots are in *task1_screenshots/*.

#	Title	Steps to reproduce	Expected vs Actual	Severity	Suspected cause
1	Whole UI renders unstyled	Open http://localhost:4100 in any browser	Expected: Conduit theme. Actual: raw HTML, no layout, no icons	Critical	index.html loads //demo.productionready.io/main.css - host is dead (404 NoSuchBucket). Fonts/icons also from external CDNs.
2	Frontend points at a dead production API	Clone repo, npm start, try to sign up	Expected: request to local/configurable API. Actual: calls conduit.productionready.io, which no longer serves the API; app unusable out of the box	Critical	API_ROOT hardcoded in src/agent.js; no env variable / .env support.
3	Error messages shown with index prefix; some errors not shown at all	Login with a wrong password; publish two articles with the same title	Expected: 'Email or password is invalid'. Actual: '0 incorrect email or password'; duplicate-title error list is empty	High	API contract mismatch - see Root Cause Analysis below.
4	1-character password accepted	Sign up with password 'x'	Expected: validation error (min length). Actual: account created and logged in	High	No password policy on client or server (password field is a plain str in schemas/users.py).
5	Article with empty title/description/body can be published	Open /editor, click Publish with all fields empty	Expected: required-field errors. Actual: HTTP 201, article with slug "" created; /article/ shows a blank page	High	No server-side required/min-length validation; slug derived from empty title.
6	Second article with same title is rejected	Publish 'My First Post' twice	Expected: unique slug (my-first-post-2). Actual: 'article already exists'; emoji-only titles collide on empty slug	Medium	Slug = slugify(title) with no uniqueness suffix.
7	Logout does not invalidate the session token	Log in, copy jwt from localStorage, log out, call GET /api/user with old token	Expected: 401. Actual: 200 - token valid for 7 days	Medium	Stateless JWT with no revocation/denylist; stored in localStorage (XSS-readable).
8	Delete Article has no confirmation	Open own article, click Delete Article	Expected: confirm dialog / undo. Actual: deleted immediately	Medium	onClick dispatches DELETE directly.
9	No maximum length on title	Publish an article with a 5,000-char title	Expected: length error. Actual: HTTP 201, layout breaks	Low	No max_length on schema/DB column.
10	Form inputs have no labels	Inspect login / editor forms (axe / DOM)	Expected: label or aria-label per input. Actual: 2 inputs with placeholder only	Low	Placeholder used as label - fails WCAG 1.3.1 / 4.1.2.
11	CORS allows any origin with credentials	Inspect app/main.py CORS config	Expected: explicit allowed origins. Actual: allow_origins=['*'] with allow_credentials=True	Low	Permissive default ALLOWED_HOSTS.

Passed checks (no bug): stored XSS in article body (<img onerror>, <script>) is sanitized by marked({sanitize:true}).

Root Cause Analysis - Bug #3: error messages render as '0 incorrect email or password'

Symptom. On a failed login the UI shows *0 incorrect email or password*. On a duplicate article title the request fails (HTTP 400) but nothing is shown, so the user thinks Publish did nothing.

Investigation. (1) In DevTools > Network the response body of POST /api/users/login is `{"errors": ["incorrect email or password"]}` - a **list**. (2) The RealWorld spec defines errors as an **object**: `{"errors": {"email or password": ["is invalid"]}}`. (3) src/components/ListErrors.js renders `Object.keys(errors).map(key => key + ' ' + errors[key])`. For an array, Object.keys returns the indexes ['0'], so the UI prints '0' + ' ' + message. (4) The editor uses the same ListErrors component (reducers/editor.js stores action.payload.errors), so the duplicate-title 400 hits the same list-vs-object mismatch; in my run no message was visible at all, so the user cannot tell why Publish failed.

Root cause. An API contract mismatch between backend and frontend: this backend's HTTP exception handler (app/api/errors/http_error.py) wraps `exc.detail` in a list, while the frontend was written against the spec's {field: [messages]} shape. Neither side validates the contract, and there is no contract/integration test.

Fix. Short term: make the backend handler return the spec shape (e.g. {"errors": {"body": [detail]}}). Defensive: ListErrors should handle both arrays and objects. Prevention: add a contract test (e.g. schema check of 4xx responses against the RealWorld OpenAPI spec, or the official Postman collection in CI) and one UI test asserting the visible error text for a wrong-password login.

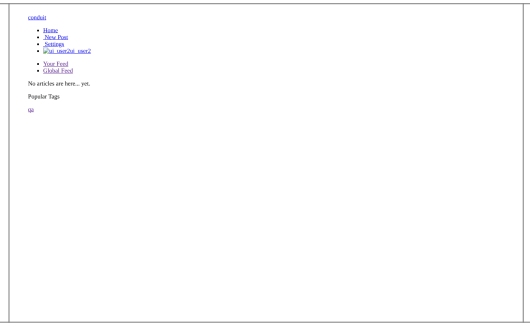
Evidence



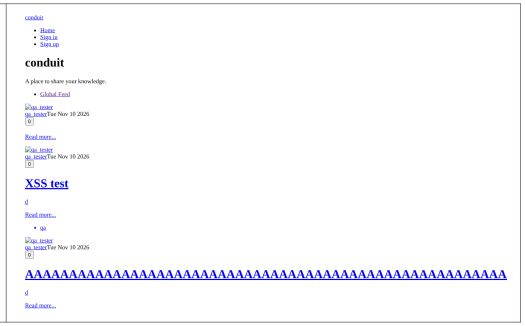
#3 wrong-password message



#5 empty article published



#4 1-char password accepted



#1 unstyled UI

Prioritisation. Fix #1-#2 first (app unusable for any new contributor), then #3-#5 (user-facing data/validation issues), then the session and UX items.